Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A coil of inductance 300mH and resistance 2Ω is connected to a source of voltage 2 V. The current reaches half of its steady state value in

(A) 0.1 s

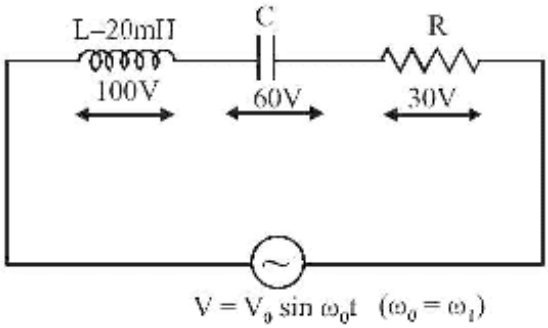
(B) 0.05 s

(C) 0.3 s

(D) 0.15 s

Q.2

Consider the RLC circuit shown below connected to an AC source of constant peak voltage V_0 and variable frequency ω_0. The value of L is 20mH. For a certain value $\omega_0 = \omega_1$, rms voltage across $\text{L}, \text{C}, \text{R}$ are shown in the diagram. At $\omega_0 = \omega_2$ it is found that rms voltage across resistance is 50 V. Then



The value of ω_2 is

(A) $\sqrt{\frac{3}{5}}\omega_1$

(B) $\sqrt{\frac{5}{3}}\omega_1$

(C) $\frac{5}{3}\omega_1$

(D) $\frac{3}{5}\omega_1$

Q.3

An ac source of angular frequency ω is fed across a resistor R and a capacitor C in series. The current in the circuit is I. If now the frequency of source is changed to $\omega/3$ (but maintaining the same voltage), the current in the circuit is found to be halved. The ratio of reactance to resistance at the original frequency ω is

(A) $\sqrt{\frac{3}{5}}$

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(B) $\sqrt{\frac{2}{5}}$

(C) $\sqrt{\frac{1}{5}}$

(D) $\sqrt{\frac{4}{5}}$

Q.4

An ideal coil of $10\mathrm{H}$ is connected in series with a resistance of 5Ω and a battery of $5\ \mathrm{V}$. 2 second after the connection is made, the current flowing in ampere in the circuit is

(A) $\left(1-e^{-1}\right)$

(B) $(1-e)$

(C) e

(D) e^{-1}

Q.5

In an oscillating LC circuit the maximum charge on the capacitor is Q. The charge on the capacitor when the energy is stored equally between the electric and magnetic field is

(A) $\frac{Q}{2}$

(B) $\frac{Q}{\sqrt{3}}$

(C) $\frac{Q}{\sqrt{2}}$

(D) Q

Q.6

The tuning circuit of a radio receiver has a resistance of 50Ω, an inductor of $10\mathrm{mH}$ and a variable capacitor. A 1 MHz radio wave produces a potential difference of $0.1\mathrm{mV}$. The value of the capacitor to produce resonance is (Take $\pi^2=10$)

(A) $2.5\mathrm{pF}$

(B) $5.0\mathrm{pF}$

(C) $25\mathrm{pF}$

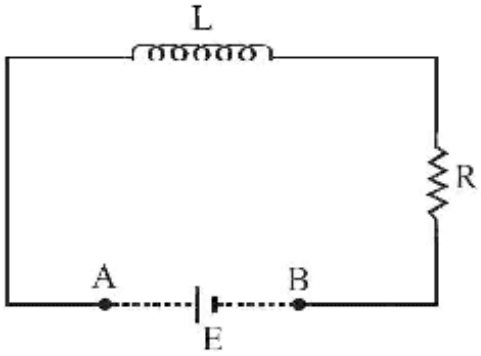
(D) $50\mathrm{pF}$

Q.7

An inductor $(L=100\mathrm{mH})$, a resistor $(R=100\Omega)$ and a battery $(E=100\ \mathrm{V})$ are initially connected in series as shown in the figure. After a long time the battery is disconnected after short circuiting the points A and

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B. The current in the circuit 1 ms after the short circuit is



(A) $1/\mathrm{eA}$

(B) eA

(C) 0.1 A

(D) 1 A

Q.8

The current in a LR circuit builds up to $\frac{3}{4}th$ of its steady state value in 4 s. The time constant of this circuit is

(A) $\frac{1}{\ell n2}s$

(B) $\frac{2}{\ell n2}s$

(C) $\frac{3}{\ell n2}s$

(D) $\frac{4}{\ell n2}s$

Q.9

An LCR series circuit is connected to a source of alternating current. At resonance, the applied voltage and the current flowing through the circuit will have a phase difference of

(A) π

(B) $\frac{\pi}{2}$

(C) $\frac{\pi}{4}$

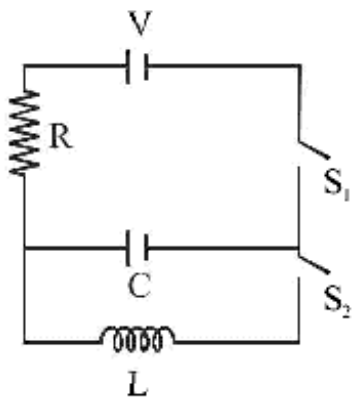
(D) 0

Q.10

In an LCR circuit as shown below both switches are open initially. Now switch S_1 is closed, S_2 kept open. (q is

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charge on the capacitor and $\tau = RC$ is Capacitive time constant). Which of the following statement is correct?



(A) Work done by the battery is half of the energy dissipated in the resistor

(B) At, $t = \tau, q = CV/2$

(C) At, $t = 2\tau, q = CV\left(1 - e^{-2}\right)$

(D) At, $t = 2\tau, q = CV\left(1 - e^{-1}\right)$

Q.11

In an alternating current circuit in which an inductance and capacitance are joined in series, current is found to be maximum when the value of inductance is 0.5 henry and the value of capacitance is $8\mu F$. Then what will be the angular frequency (in rad/s) of applied alternating voltage?

(A) 490 rad/s

(B) 495 rad/s

(C) 505 rad/s

(D) 500 rad/s

Q.12

If an LCR series circuit is connected to an ac source, then at resonance the

(A) voltage across R is zero

(B) voltage across R equals the applied voltage

(C) voltage across C is zero

(D) voltage across L equals the applied voltage

Q.13

Of the following about capacitive reactance which one is correct?

(A) The reactance of the capacitor is directly proportional to its ability to store charge

(B) Capacitive reactance is inversely proportional to the frequency of the current

(C) Capacitive reactance is measured in farad

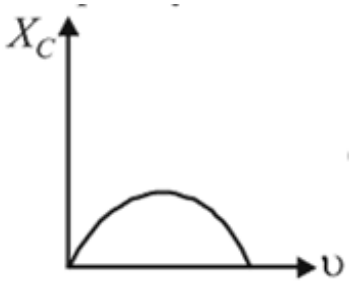
(D) The reactance of a capacitor in an A.C. circuit is similar to the resistance of a capacitor in a D.C. circuit

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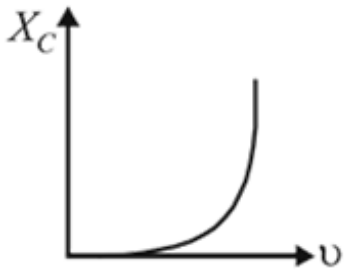
Q.14

Which of the following graphs represents the correct variation of capacitive reactance X_C with frequency U?

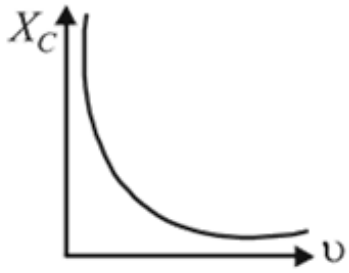
(A)



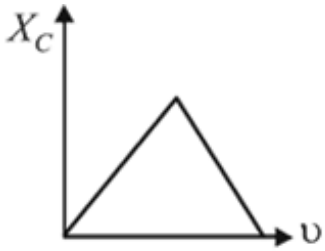
(B)



(C)



(D)



Q.15

In a series LCR circuit, the difference of the frequencies at which current amplitude falls to $\frac{1}{\sqrt{2}}$ of the current amplitude at resonance is

(A) $\frac{R}{2\pi L}$

(B) $\frac{R}{\pi L}$

(C) $\frac{2R}{\pi L}$

(D) $\frac{3R}{2\pi L}$

Q.16

A coil of inductance 8.4mH and resistance 6Ω is connected to a 12 V battery. The current in the coil is 1.0 A at approximately the time

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(A) $500\ \mathrm{s}$

(B) $25\ \mathrm{s}$

(C) $35\ \mathrm{ms}$

(D) $1\ \mathrm{ms}$

Q.17

In an $L-R$ circuit, the value of L is $(0.4/\pi)$ henry and the value of R is $30\ \mathrm{ohm}$. If in the circuit, an alternating emf of 200 volt at 50 cycles per second is connected, the impedance of the circuit and current will be:

(A) $11.4\ \mathrm{ohm}, 17.5$ ampere

(B) $30.7\ \mathrm{ohm}, 6.5$ ampere

(C) $40.4\ \mathrm{ohm}, 5$ ampere

(D) $50\ \mathrm{ohm}, 4$ ampere.

Q.18

An ac voltage is applied to a resistance R and an inductor L in series. If R and the inductive reactance are both equal to 3Ω, the phase difference between the applied voltage and the current in the circuit is

(A) $\pi/6$

(B) $\pi/4$

(C) $\pi/2$

(D) zero

Q.19

A series R-C circuit is connected to an alternating voltage source. Consider two situations:
(A) When capacitor is air filled.
(B) When capacitor is mica filled. Current through resistor is I and voltage across capacitor is V then:

(A) $\mathrm{V_a} > \mathrm{V_b}$

(B) $\mathrm{I_a} > \mathrm{I_b}$

(C) $\mathrm{V_a} = \mathrm{V_b}$

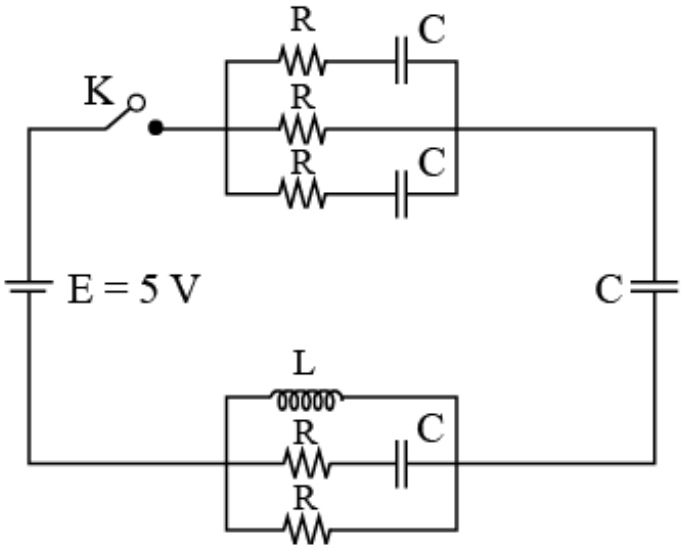
(D) $\mathrm{V_a} < \mathrm{V_b}$

Q.20

Find the current passing through battery immediately after key (K) is closed. It is given that initially all the

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capacitors are uncharged. (given that $\mathrm{R} = 6\Omega$ and $\mathrm{C} = 4\mu\mathrm{F}$)



(A) 1 A

(B) 5 A

(C) 3 A

(D) 2 A

Answers & Solutions

Q.1 Answer:

0.1 s

Solution:

The charging of inductance given by, $i = i_0\left(1 - e^{-\frac{Rt}{L}}\right)$

$\frac{i_0}{2} = i_0\left(1 - e^{-\frac{Rt}{L}}\right) \Rightarrow \mathrm{e}^{-\frac{\mathrm{Rt}}{\mathrm{L}}} = \frac{1}{2}$

Taking log on both the sides, $-\frac{Rt}{L} = \log 1 - \log 2$

$\Rightarrow t = \frac{L}{R}\log 2 = \frac{300\times10^{-3}}{2} \times 0.69$

$\Rightarrow t = 0.1\mathrm{sec}$

Q.2 Answer:

$\sqrt{\frac{3}{5}}\omega_1$

Solution:

If voltage across resistor is 50 V **then this should be the resonance condition. At resonance,** $X_L = X_C$ $\omega_2\,\mathrm{L} = \frac{1}{\omega_2 \mathrm{C}}; \omega_2 = \frac{1}{\sqrt{\mathrm{LC}}}$

Also, At $\omega = \omega_1$

$I = \frac{100}{X_L} = \frac{60}{X_C}; \frac{100}{\omega_1 L} = \frac{60}{1/\omega_1 C}$

$\mathrm{C} = \frac{100}{\omega_1^2\,\mathrm{L}\times 60} = \frac{5}{3\omega_1^2\,\mathrm{L}}$

$\omega_2 = \frac{1}{\sqrt{L\times\frac{5}{3\omega_1^2 L}}} = \sqrt{\frac{3}{5}}\omega_1$

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Q.3 Answer:

$\sqrt{\frac{3}{5}}$

Solution:

At angular frequency ω, the current in RC circuit is given by

$i_{\max}=\frac{\mathrm{V}_{\max}}{\sqrt{\mathrm{R}^2+\left(\frac{1}{\omega \mathrm{C}}\right)^2}} \ldots \ldots .(\mathrm{i})$

Also $\frac{\mathrm{i}_{\mathrm{rms}}}{2}=\frac{\mathrm{V}_{\mathrm{rms}}}{\sqrt{\mathrm{R}^2+\left(\frac{1}{\frac{\omega}{3} \mathrm{C}}\right)^2}}=\frac{\mathrm{V}_{\max}}{\sqrt{\mathrm{R}^2+\frac{9}{\omega^2 \mathrm{C}^2}}} \ldots \ldots .(\mathrm{ii})$

From equation (i) and (ii) we get

$3 \mathrm{R}^2=\frac{5}{\omega^2 \mathrm{C}^2} \Rightarrow \frac{\frac{1}{\omega \mathrm{C}}}{\mathrm{R}}=\sqrt{\frac{3}{5}} \Rightarrow \frac{\mathrm{X}_{\mathrm{C}}}{\mathrm{R}}=\sqrt{\frac{3}{5}}$

Q.4 Answer:

$\left(1-e^{-1}\right)$

Solution:

$I=I_o\left(1-e^{-\frac{R}{L}t}\right)$

(When current is in growth in LR circuit)

$=\frac{E}{R}\left(1-e^{-\frac{R}{L}t}\right)=\frac{5}{5}\left(1-\mathrm{e}^{-\frac{5}{10} \times 2}\right)$

$=\left(1-e^{-1}\right)$

Q.5 Answer:

$\frac{Q}{\sqrt{2}}$

Solution:

When the capacitor is completely charged, the total energy in the LC circuit is with the capacitor and that

energy is $E=\frac{1}{2}\frac{Q^2}{C}$

When half energy is with the capacitor in the form of electric field between the plates of the capacitor we get

$\frac{E}{2}=\frac{1}{2}\frac{Q'^2}{C}$ **where Q ' is the charge on one plate of the capacitor**

$\therefore \frac{1}{2} \times \frac{1}{2}\frac{Q^2}{C}=\frac{1}{2}\frac{Q'^2}{C} \Rightarrow Q'=\frac{Q}{\sqrt{2}}$

Q.6 Answer:

$2.5 \mathrm{pF}$

Solution:

$\mathrm{L}=10 \mathrm{mHz}=10^{-2} \mathrm{~Hz}$

$\mathrm{f}=1 \mathrm{MHz}=10^6 \mathrm{~Hz}$

$\mathrm{f}=\frac{1}{2\pi\sqrt{\mathrm{LC}}}$

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$f^2 = \frac{1}{4\pi^2 LC}$
$\Rightarrow C = \frac{1}{4\pi^2 f^2 L} = \frac{1}{4\times 10\times 10^{-2}\times 10^{12}} = \frac{10^{-12}}{0.4} = 2.5\text{pF}$

Q.7 Answer:

$1/eA$

Solution:

Initially, when steady state is achieved,
$i = \frac{E}{R}$
Let E is short circuited at $t = 0$. Then
At $t = 0, i_0 = \frac{E}{R}$
Let during decay of current at any time the current flowing is $-L\frac{di}{dt} - iR = 0$
$\Rightarrow \frac{di}{i} = -\frac{R}{L}dt \Rightarrow \int_{i_0}^{i}\frac{di}{i} = \int_0^t -\frac{R}{L}dt$
$\Rightarrow \log_e \frac{i}{i_0} = -\frac{R}{L}t \Rightarrow i = i_0 e^{-\frac{R}{L}t}$
$\Rightarrow i = \frac{E}{R}e^{-\frac{R}{L}t} = \frac{100}{100}e^{\frac{-100\times 10^{-3}}{100\times 10^{-3}}} = \frac{1}{e}$

Q.8 Answer:

$\frac{2}{\ell n 2}s$

Solution:

We know that, $i = i_0\left(1 - e^{-t/\tau}\right)$
or $\frac{3}{4}i_0 = i_0\left(1 - e^{-4/\tau}\right)$
or $e^{-4/\tau} = \frac{1}{4}$
or $e^{4/\tau} = 4$
$\therefore \quad \frac{4}{\tau} = \ln 4$
or $\quad \tau = \frac{2}{\ln 2}\ \text{s}$

Q.9 Answer:

0

Solution:

At resonance, $\omega\text{L} = \frac{1}{\omega\text{C}}$. The circuit behaves as if it contains R only. So, phase difference $= 0$ At resonance, impedance is minimum $Z_{\min} = R$ and current is maximum, given by $\mathbf{I}_{\max} = \frac{\text{E}}{\text{Z}_{\min}} = \frac{\text{E}}{\text{R}}$
It is interesting to note that before resonance the current leads the applied emf, at resonance it is in phase, and after resonance it lags behind the emf. LCR series circuit is also called as acceptor circuit and parallel LCR circuit is called rejector circuit.

Q.10 Answer:

$\text{At, t} = 2\tau, \text{q} = \text{CV}\left(1 - \text{e}^{-2}\right)$

Solution:

Charge on he capacitor at any time t is given by $q = CV\left(1 - \text{e}^{\text{t}/\tau}\right)$
at $\text{t} = 2\tau$

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$q = CV\left(1 - e^{-2}\right)$

Q.11 Answer:

500 rad/s

Solution:

Current is maximum when $\mathrm{X_L} = \mathrm{X_C}$
$\Rightarrow \omega \mathrm{L} = \frac{1}{\omega \mathrm{C}} \Rightarrow \omega = \frac{1}{\sqrt{\mathrm{LC}}} = \frac{1}{\sqrt{0.5 \times 8 \times 10^{-6}}}$
$= \frac{1}{2 \times 10^{-3}} = 500 \mathrm{rad/s}$

Q.12 Answer:

voltage across R equals the applied voltage

Solution:

In series RLC circuit,
Voltage, $\mathrm{V} = \sqrt{\mathrm{V_R^2} + \left(\mathrm{V_L} - \mathrm{V_C}\right)^2}$
And, at resonance, $V_L = V_C$
Hence, $\mathrm{V} = \mathrm{V_R}$

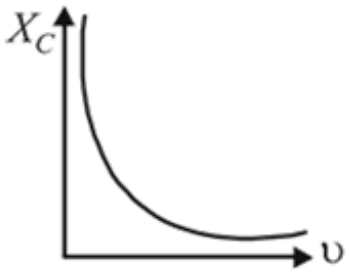
Q.13 Answer:

Capacitive reactance is inversely proportional to the frequency of the current

Solution:

$\mathrm{X_C} = \frac{1}{\omega \mathrm{C}} \Rightarrow \mathrm{X_C} \alpha \frac{1}{\omega}$ **for given** C

Q.14 Answer:



Solution:

Capacitive reactance, $X_C = \frac{1}{\omega C} = \frac{1}{2\pi v C}$
$\Rightarrow \quad X_L \propto \frac{1}{v}$
With increases in frequency, X_C **decreases. Hence, option (c) represents the corrent graph.**

Q.15 Answer:

$\frac{R}{2\pi L}$

Solution:

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At resonance $I_R = \frac{E_0}{R} \Rightarrow \frac{I_R}{\sqrt{2}} = \frac{E_0}{\sqrt{2}R} = \frac{E_0}{\sqrt{R^2+\left(\omega L - \frac{1}{\omega C}\right)^2}}$

$\Rightarrow \omega_1 L - \frac{1}{\omega_1 C} = -R$ **and** $\omega_2 L - \frac{1}{\omega_2 C} = +R$

$\Rightarrow L(\omega_1 + \omega_2) = \left(\frac{\omega_1+\omega_2}{\omega_1\omega_2}\right)\frac{1}{C} \Rightarrow \omega_1\omega_2 = \frac{1}{LC}$

and $L(\omega_2 - \omega_1) + \left(\frac{\omega_2-\omega_1}{\omega_1\omega_2}\right)\frac{1}{C} = 2R$

Putting value of ω_1 **and** ω_2 **we get**

$\omega_2 - \omega_1 = \frac{R}{L} \Rightarrow f_2 - f_1 = \frac{R}{2\pi L}$

Q.16 Answer:

1 ms

Solution:

Using $I = I_0\left(1 - e^{-t/\tau}\right)$

But $I_0 = \frac{V}{R}$ **and** $\tau = \frac{L}{R}$

$\therefore \quad I = \frac{V}{R}\left(1 - e^{-Rt/L}\right) = \frac{12}{6}\left[1 - e^{-6t/8.4\times10^{-3}}\right.$

$= 1$ (given)

$t = 0.97 \times 10^{-3}$ s ≈ 1 ms

Q.17 Answer:

50 ohm, 4 ampere.

Solution:

Here, $X_L = \omega L = 2\pi f L = 2\pi \times 50 \times \frac{0.4}{\pi} = 40\Omega$

$R = 30\Omega$

$Z = \sqrt{R^2 + X_L^2} = \sqrt{30^2 + 40^2} = 50\Omega$

$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{200}{50} = 4$ A

Q.18 Answer:

$\pi/4$

Solution:

The phase difference ϕ **is given by** $\tan\phi = \frac{X_L}{R} = \frac{3}{3} = 1 \Rightarrow \phi = \frac{\pi}{4}$

Q.19 Answer:

$\mathrm{V_a} > \mathrm{V_b}$

Solution:

For series $\mathrm{C-R}$ **circuit, the impedance** $\mathrm{Z} = \sqrt{\mathrm{R}^2 + \mathrm{X_C^2}}$ **where** $\mathrm{X_C} = \frac{\mathrm{j}}{\omega \mathrm{C}}$ **and current** $\mathrm{I} = \mathrm{V}/\mathrm{Z}$
When the capacitor is filled by mica, the capacitance will be increased. If C **increases,** X_C **decreases, so the current will increase and hence voltage across resistance increases and voltage across capacitor decreases. thus,** $\mathrm{V_a} > \mathrm{V_b}$

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Q.20 Answer:

1 A

Solution:

$R_{\text{eq}} = \frac{5R}{6} \Rightarrow I = \frac{6E}{5R} = 1A$

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